

# The effect of two calcium silicate-based and one epoxy resin-based root canal sealer on postoperative pain: a randomized controlled trial

T. Aslan<sup>1</sup>  & H. Dönmez Özkan<sup>2</sup> 

<sup>1</sup>Department of Endodontics, Faculty of Dentistry, Erciyes University, Kayseri; and <sup>2</sup>Department of Endodontics, Faculty of Dentistry, Aydın Adnan Menderes University, Efeler, Turkey

## Introduction

Postoperative pain arising after root canal treatment is undesirable but may manifest in hours and may persist for days following treatment (Wang *et al.* 2010). The incidence of post-treatment endodontic pain has been reported to be between 3 and 58% (Sathorn *et al.* 2008, Gondim *et al.* 2010) with severe pain being observed at a rate of less than 12% of the

entire population after treatment (Ng *et al.* 2004). Acute inflammation was thought to be one of the main reasons for post-treatment endodontic severe pain, that occurs following chemical, mechanical or microbial injuries in periapical tissue (Seltzer & Naird 2004). Inappropriate instrumentation or filling of the root canals has been ascribed as one of the factors predisposing to postoperative pain (Goreva & Petrikas 2004). It has been reported that extrusion of the

Correspondence: Tuğrul Aslan, Department of Endodontics, Faculty of Dentistry, Erciyes University, Prof. Dr. N. Taker Street, Melikgazi, Kayseri 38039, Turkey (Tel.: +90-506-810-7688; e-mail: dr.tugrulaslan@hotmail.com).

sealer during root canal filling has cytotoxic effects on periapical tissues, causing periapical inflammation, necrosis and pain (Froes *et al.* 2009). The chemical composition of sealers also affects the degree of tissue reaction (Zhang & Peng 2015).

Calcium silicate-based endodontic sealers promote periapical tissue healing by reducing inflammation and inducing the differentiation of odontoblasts and mineralization in preosteoblasts (Lee *et al.* 2019). In an *in vitro* study, calcium silicate-based endodontic sealers were reported to induce less severe cytotoxic effects compared to their resin-based counterpart AH Plus (Dentsply DeTrey GmbH, Konstanz, Germany; Zhang *et al.* 2010). In contrast, in a recent randomized clinical trial, the use of a bioceramic sealer or an epoxy resin sealer (AH Plus) during root canal treatment was reported to lead to similar postoperative pain in teeth with acute apical periodontitis (Grau-naite *et al.* 2018).

EndoSeal MTA (Maruchi, Wonju, Korea) has been marketed as a pozzolan-based MTA sealer. Previous reports have demonstrated that this injectable biotight root canal sealer provides satisfactory biological and physical properties (Seo *et al.* 2019) and adequate bond strength to dentine (Silva *et al.* 2016). However, to date, the effect of this sealer on postoperative pain has never been investigated. Furthermore, there are few studies in the literature regarding the potential effects of calcium silicate-based root canal sealer on postoperative pain. Therefore, the present study aimed to evaluate the effect of two calcium silicate-based root canal sealers, Endoseal MTA and EndoSequence BC Sealer, on postoperative pain following single-visit root canal treatment on molar teeth compared to their epoxy/amine resin-based counterpart AH Plus. The null hypotheses tested in this study were as follows:

1. The type of sealer used would not change the incidence and the intensity of post-treatment endodontic pain
2. The analgesic intake of patients following single-visit root canal treatment would not differ amongst the experimental groups.

## Materials and methods

The Ethics Committee of Erciyes University of Medical Sciences (Decision no: 2018/08) approved this randomized prospective clinical trial. The study protocol was recorded in [www.ClinicalTrials.gov](http://www.ClinicalTrials.gov) databases, with the NCT number (NCT/03919318). The required minimum sample size was calculated using the

G\*Power v.3.1.9.2 program (Heinrich Heine, Düsseldorf University, Düsseldorf, Germany) according to the data of a previous study (Alonso-Ezpeleta *et al.* 2012). An alpha-type error of 0.05 and a beta power of 0.85 was specified, and the minimal estimated sample size for each group was computed as 24 to determine the postoperative pain differences between three study groups. To increase the statistical power of this study and considering potential patient dropouts, it was decided to enrol 32 patients per group. Therefore, a total of 96 patients were included in the study.

The patients were selected from those who were referred to the Hospitals of Dental Faculty of Erciyes University, Kayseri, Turkey and Dental Faculty of Adnan Menderes University, Aydın, Turkey for root canal treatment from February 2018 to July 2019. Only patients who had mandibular first and second molar teeth diagnosed with asymptomatic irreversible pulpitis due to deep caries were included. The clinical diagnosis of asymptomatic irreversible pulpitis was based on the presence of a deep carious lesion that would cause a large pulp exposure during its removal and in the absence of clinical and radiographic pathosis and symptoms. Additional identifiers for this diagnosis were that after exposing the pulp, profuse bleeding of the pulp having a thick consistency and an inability to achieve haemostasis within 2–3 min. Inclusion and exclusion criteria are listed in Table 1. All participants were informed accordingly and signed a written informed consent form.

## Treatment procedures

All procedures were performed by two specialists having at least 10 years of clinical experience in the university clinics. Throughout the entire procedure, the operators used dental operating loupes at  $\times 4$  magnification. All patients received a single-visit root canal treatment to limit the potential pain-inducing factors that might be caused by multiple visits. During the diagnostic examination, thermal tests (EndoIce; Coltene/Whaledent Inc, Altstätten, Switzerland) and electric pulp tests (Parkell, Edgewood, NY, USA) were performed to determine pulp sensibility.

Inferior alveolar nerve block anaesthesia (4% articaine hydrochloride with 1 : 100 000 adrenalin) was performed. None of the patients required supplementary anaesthesia. Then, the related tooth was isolated with rubber dam, caries and existing restorations were removed with high-speed diamond burs under water cooling. The vitality of the pulp was confirmed after the

**Table 1** Inclusion/exclusion criteria of the participants

| Inclusion criteria                                                                                                                        | Exclusion criteria                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Participants between 18 and 60 years of ages                                                                                              | Patients who refuse to participate this study                                                                                                |
| Good oral hygiene                                                                                                                         | Medically compromised patients (with immunosuppressive/systemic diseases, patients on medications)                                           |
| Patients had not used any analgesic in last 7 days                                                                                        | Symptomatic or devital teeth                                                                                                                 |
| Patients had not used any antibiotic in last 7 days                                                                                       | The presence of advanced periodontal disease (probing depth >4 mm)                                                                           |
| Prolonged positive response to cold test (Endolce; Coltene/Whaledent Inc, Cuyahoga Falls, OH) and electric pulp tester (Parkell, NY, USA) | The presence of open apex, presence of calcification, presence of resorption                                                                 |
| Patients diagnosed with asymptomatic irreversible pulpitis caused by deep carious lesion on the mandibular first or second molar teeth    | Patients who had multiple teeth requiring endodontic treatment                                                                               |
| The presence of profusely pulp bleeding with a thick consistency, which is exposed during caries removing                                 | Patients with allergic sensitivity to materials and agents that should be used during the root canal treatment                               |
| Patients who had healthy periapical tissues (confirmed with periapical radiography)                                                       | Patients who had allergic sensitivity to local anaesthetics                                                                                  |
|                                                                                                                                           | Patients who had systemic or allergic sensitivity for the NSAIDs                                                                             |
|                                                                                                                                           | Pregnant patients and patients in lactation period                                                                                           |
|                                                                                                                                           | Overfilling (extrusion of the gutta-percha or sealer beyond the radiographic apex) or short filling (>2 mm short from the radiographic apex) |
|                                                                                                                                           | The teeth with extensive coronal destruction that need a core build-up                                                                       |

visualization of a bleeding in the pulp chamber. Next, the root canal length determination was performed using Raypex 6 apex locator (VDW, Munich, Germany) with a size 8 or 10 K-file (NitiFlex; Dentsply Sirona, Ballaigues, Switzerland) and confirmed radiographically. Apical patency was established and maintained with size 10 K-file. Initial canal shaping was performed up to a size 20 K-file, and then, a nickel–titanium file system Reciproc (VDW) R25 files (size 25, 0.08 taper) at the working length used with a VDW Reciproc Silver (VDW) endodontic motor in the “RECIPROC ALL” mode. When necessary, additional instrumentation was performed with R40 files (size 40, 0.06 taper) according to the initial apical diameter. A total of 10 mL of 5% NaOCl were used with a 30-gauge irrigation needle (Max-i-Probe; Dentsply Sirona). In all irrigation processes, the irrigation needle was introduced into the canal 1.5 mm shorter than the working length, and the irrigant delivered with moderate pressure to avoid an irrigant accident. Finally, 3 mL of 17% EDTA, 3 mL of 5% NaOCl and 2 mL of distilled water were used, respectively, for final irrigation. During the final NaOCl irrigation process, manual dynamic irrigation was performed with the push-pull motion of a well-fitting gutta-percha cone for 30 s. Finally, the root canals were dried with paper points. Thereafter,

participants were randomly allocated to a group using a computer algorithm program (<http://random.org>;  $n = 32$ ). Allocation was done by a trained dental assistant who was blinded to the study procedures to prevent bias.

Group 1: AH Plus (Dentsply).

Group 2: Endoseal MTA (Maruchi).

Group 3: EndoSequence BC Sealer (Brasseler).

After the allocation of patients into the experimental groups, six patients stated that they wanted to withdraw from the study at the beginning of the canal filling process; therefore, they were excluded from the study. Since the remaining 90 patients wished to continue, 30 patients were allocated into each experimental group ( $n = 30$ ).

A single tapered gutta-percha cone [Reciproc R25 or Reciproc R40 gutta-percha cones (VDW)] was adapted to the root canal, and the position of the cone was confirmed with a periapical radiograph. Afterwards, the procedure described by Castelo-Baz *et al.* (2013) was utilized for sealer application with suitable paper point cones (1 mm shorter than working length) as follows: the first paper point was used to apply the sealer, the second one to distribute and the third one used to remove excess sealer. Following

sealer application, the root canal was filled with a single cone placed in the canal. Subsequently, excess gutta-percha cone was removed and the pulp chamber was cleaned. Then, coronal access cavities were restored with a direct adhesive build-up using a composite resin material (Single Bond Universal; 3M ESPE, St. Paul, MN, USA). Finally, ibuprofen 400 mg was prescribed and the patients were recommended to use it only when they encountered severe pain as previously suggested (Keskin *et al.* 2017).

### **Evaluation of postoperative pain**

After treatment, two forms were given to the participants. The first form designed to record pain levels was based on the Visual Analog Scale (VAS), where VAS pain levels were indicated on the chart by the participants. The value "0" was defined as "no pain," and "100" was defined as "unbearable pain."

In the second form, participants were asked to report the frequency of analgesic drug intake. The patients were asked to choose one of the three options: "0: No pain, or pain which does not require the use of analgesics," "1: Moderate pain which can be well controlled by the use of analgesics and does not affect daily activities or sleep," "2: Impaired daily activities due to unbearable pain that cannot be controlled by the use of analgesics." Participants were asked to complete these two forms at 6, 12, 24 and 48 h after treatment and on the 3rd, 4th, 5th, 6th and 7th days. A phone call was made every day for 7 days to obtain information on the postoperative pain and the frequency of analgesic intake. The patients were asked to call the contact number on the form if they encountered severe pain or if they needed to ask any questions regarding the treatment.







- Camargo CH, Camargo SE, Valera MC, Hiller KA, Schmalz G, Schweikl H (2009) The induction of cytotoxicity, oxidative stress, and genotoxicity by root canal sealers in mammalian cells. *Oral Surgery Oral Medicine Oral Pathology Oral Radiology and Endodontics* **108**, 952–60.
- Castelo-Baz P, Martin-Biedma B, Lopes MM *et al.* (2013) Ultramicroscopic study of the interface and sealing ability of four root canal obturation methods: resilon versus gutta-percha. *Australian Endodontic Journal* **39**, 159–63.
- Comparin D, Moreira EJJ, Souza EM, De-Deus G, Arias A, Silva E (2017) Postoperative pain after endodontic retreatment using rotary or reciprocating instruments: a randomized clinical trial. *Journal of Endodontics* **43**, 1084–8.
- Froes FG, Miranda AM, Abad Eda C, Riche FN, Pires FR (2009) Non-surgical management of paraesthesia and pain associated with endodontic sealer extrusion into the mandibular canal. *Australian Endodontic Journal* **35**, 183–6.
- Gondim E Jr, Setzer FC, Dos Carmo CB, Kim S (2010) Postoperative pain after the application of two different irrigation devices in a prospective randomized clinical trial. *Journal of Endodontics* **36**, 1295–301.
- Goreva LA, Petrikas AZ (2004) Postobturation pain associated with endodontic treatment. *Stomatologia* **83**, 14–6.
- Graunaite I, Skuicaite N, Lodiene G, Agentiene I, Machiulskiene V (2018) Effect of resin-based and bioceramic root canal sealers on postoperative pain: a split-mouth randomized controlled trial. *Journal of Endodontics* **44**, 689–93.
- Keskin C, Özdemir Ö, Uzun İ, Güler B (2017) Effect of intracanal cryotherapy on pain after single-visit root canal treatment. *Australian Endodontic Journal* **43**, 83–8.
- Lee BN, Hong JU, Kim SM *et al.* (2019) Anti-inflammatory and osteogenic effects of calcium silicate-based root canal sealers. *Journal of Endodontics* **45**, 73–8.
- Lodiene G, Morisbak E, Bruzell E, Ørstavik D (2008) Toxicity evaluation of root canal sealers in vitro. *International Endodontic Journal* **41**, 72–7.
- Mehrvarzfar P, Abbott PV, Saghi MA *et al.* (2012) Effects of three oral analgesics on postoperative pain following root canal preparation: a controlled clinical trial. *International Endodontic Journal* **45**, 76–82.
- Morse DR, Furst ML, Belott RM, Lefkowitz RD, Spritzer IB, Sideman BH (1987) Infectious flare-ups and serious sequelae following endodontic treatment: a prospective randomized trial on efficacy of antibiotic prophylaxis in cases of asymptomatic pulpal-periapical lesions. *Oral Surgery Oral Medicine and Oral Pathology* **64**, 96–109.
- Morse DR, Esposito JV, Furst ML (1990) Comparison of prophylactic and on-demand diflunisal for pain management of patients having one-visit endodontic therapy. *Oral Surgery Oral Medicine and Oral Pathology* **69**, 729–36.
- Mulhern JM, Patterson SS, Newton CW, Ringel AM (1982) Incidence of postoperative pain after one-appointment endodontic treatment of asymptomatic pulpal necrosis in single-rooted teeth. *Journal of Endodontics* **8**, 370–5.
- Ng YL, Glennon JP, Setchell DJ, Gulabivala K (2004) Prevalence of and factors affecting post-obturation pain in patients undergoing root canal treatment. *International Endodontic Journal* **37**, 381–91.
- Ng YL, Mann V, Gulabivala K (2011) A prospective study of the factors affecting outcomes of nonsurgical root canal treatment: part 1: periapical health. *International Endodontic Journal* **44**, 583–609.
- Okiji T, Yoshida K (2009) Reparative dentinogenesis induced by mineral trioxide aggregate: a review from the biological and physicochemical points of view. *International Journal of Dentistry*. <https://doi.org/10.1155/2009/464280>
- Omoigui S (2007) The biochemical origin of pain: the origin of all pain is inflammation and the inflammatory response. Part 2 of 3 - inflammatory profile of pain syndromes. *Medical Hypotheses* **69**, 1169–78.
- Sari S, Duruturk L (2007) Radiographic evaluation of periapical healing of permanent teeth with periapical lesions after extrusion of AH Plus sealer. *Oral Surgery, Oral Medicine, Oral Pathology, Oral Radiology, and Endodontology* **104**, 54–9.
- Sathorn C, Parashop P, Messer H (2008) The prevalence of postoperative pain and flare-up in single and multiple-visit endodontic treatment: a systematic review. *International Endodontic Journal* **41**, 91–9.
- Seo DG, Lee D, Kim YM, Song D, Kim SY (2019) Biocompatibility and mineralization activity of three calcium silicate-based root canal sealers compared to conventional resin-based sealer in human dental pulp stem cells. *Materials* **12**, 2482.
- Seltzer S, Naidorf IJ (2004) Flare-ups in endodontics: I. Etiological factors. *Journal of Endodontics* **30**, 476–81.
- Silva EJ, Carvalho NK, Prado MC *et al.* (2016) Push-out bond strength of injectable pozzolan-based root canal sealer. *Journal of Endodontics* **42**, 1656–9.
- Vengerfeldt V, Mändar R, Saag M, Piir A, Kullisaar T (2017) Oxidative stress in patients with endodontic pathologies. *Journal of Pain Research* **10**, 2031–40.
- Wang C, Xu P, Ren L, Dong G, Ye L (2010) Comparison of post-obturation pain experience following one-visit and two-visit root canal treatment on teeth with vital pulps: a randomized controlled trial. *International Endodontic Journal* **43**, 692–7.
- Zhang W, Li Z, Peng B (2010) Ex vivo cytotoxicity of a new calcium silicate-based canal filling material. *International Endodontic Journal* **43**, 769–74.
- Zhang W, Peng B (2015) Tissue reactions after subcutaneous and intraosseous implantation of iRoot SP, MTA and AH Plus. *Dental Materials Journal* **34**, 774–80.